

The prism, the parallelepiped and the cube

The family, its sections, and the diagonal formula that every cuboid problem reduces to.

LESSON 21–22

2 × 40 MIN

GEOMETRY 11, §10

IGCSE E13.X

LESSON CLOCK

12'

22'

24'

22'

16'

4

The family	12 min
Sections	22 min
The parallelepiped	24 min
The diagonal formula	22 min
Practice	16 min
Homework	4 min

LEARNING OBJECTIVES

- Name the elements of a prism and classify it as right, oblique or regular.
- Describe the sections of a prism parallel to the base and through a lateral edge.
- Prove the properties of a parallelepiped.
- Use the space-diagonal formula for a cuboid.

TEXTBOOK

National	pp. 101–112
Cambridge	Core & Extended, pp. 289–296
Workbook	Exercise 10.1

01

Explanation

The family

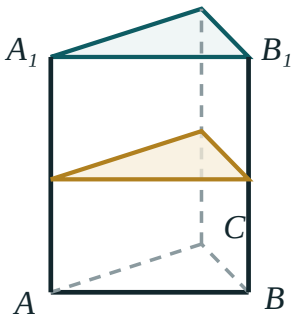
DEFINITION

A **prism** has two congruent parallel bases joined by parallelograms. It is **right** when the lateral edges are perpendicular to the bases, and **regular** when it is right and its base is a regular polygon.

SOLID	BASE	LATERAL FACES
oblique prism	any polygon	parallelograms
right prism	any polygon	rectangles
regular prism	regular polygon	congruent rectangles
parallelepiped	parallelogram	parallelograms
cuboid	rectangle	rectangles
cube	square	squares

Each row is a special case of the one above it. Every theorem proved for a prism therefore holds for a cube.

Sections



a section parallel
to the base is
congruent to it

every cross-section of a prism repeats the base
A plane parallel to the base cuts a triangle congruent to it — at every height.

CUTTING PLANE	SECTION
parallel to the base	congruent to the base
through two lateral edges	a parallelogram (a diagonal section)
perpendicular to the lateral edges	the perpendicular section

The perpendicular section matters because the lateral surface area of **any** prism, right or oblique, is the perimeter of the perpendicular section times the lateral edge.

The parallelepiped

Four properties, each proved from the parallel-plane theorems of Grade 10:

- 1. Opposite faces are congruent parallelograms lying in parallel planes.
- 2. All four space diagonals meet at one point and bisect each other.
- 3. That point is the centre of symmetry of the solid.
- 4. In a **cuboid** all four space diagonals are equal.

Property 2 says the parallelepiped is the space analogue of the parallelogram, whose diagonals also bisect each other. The proof is the same: the two diagonals of any diagonal section are the diagonals of a parallelogram.

The diagonal formula

$$d^2 = a^2 + b^2 + c^2$$

Pythagoras twice — once in the base to get $\sqrt{a^2 + b^2}$, once vertically. For a cube of edge a it gives $d = a\sqrt{3}$.

CUBOID	BASE DIAGONAL	SPACE DIAGONAL
$3 \times 4 \times 12$	5	13
$1 \times 2 \times 2$	$\sqrt{5}$	3
$2 \times 3 \times 6$	$\sqrt{13}$	7
$a \times a \times a$	$a\sqrt{2}$	$a\sqrt{3}$

THREE SQUARES, NOT TWO

Every edge contributes. Learners who use only the two base edges get the **face** diagonal and answer a different question.

02 Worked examples

EXAMPLE 1 A cuboid measures $6 \times 8 \times 10$. Find its space diagonal and its four diagonal sections.

$$① \quad d = \sqrt{36 + 64 + 100} = \sqrt{200}$$

$$② \quad = 10\sqrt{2} \approx 14.1$$

$$③ \quad \text{Diagonal sections are rectangles } 10 \times \sqrt{100} \text{ and } 10 \times \sqrt{136}.$$

Two different sizes.

ANSWER

$$10\sqrt{2} \approx 14.1$$

EXAMPLE 2 A regular hexagonal prism has base edge 4 and height 9. Find the longest diagonal.

$$① \quad \text{The longest base diagonal of a regular hexagon is } 2 \times 4 = 8.$$

$$② \quad d = \sqrt{64 + 81}$$

$$③ \quad = \sqrt{145} \approx 12.04$$

ANSWER

$$\sqrt{145} \approx 12.0$$

EXAMPLE 3 A cube has space diagonal $9\sqrt{3}$. Find its edge and its face diagonal.

① $a\sqrt{3} = 9\sqrt{3}$
 $a = 9$

② Face diagonal $a\sqrt{2} = 9\sqrt{2}$

③ ≈ 12.7

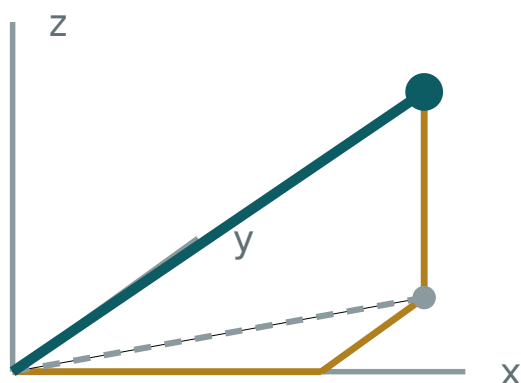
ANSWER

Edge 9, face diagonal $9\sqrt{2}$

03 Interactive model

Hold a box and trace the base diagonal, then the space diagonal, saying the two Pythagoras steps.

The diagonal of a box



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Prism	Prizma	Призма
Right prism	To'g'ri prizma	Прямая призма
Oblique prism	Og'ma prizma	Наклонная призма
Regular prism	Muntazam prizma	Правильная призма
Parallelepiped	Parallelepiped	Параллелепипед
Cuboid	To'g'ri burchakli parallelepiped	Прямоугольный параллелепипед
Lateral edge	Yon qirra	Боковое ребро
Diagonal section	Diagonal kesim	Диагональное сечение
Space diagonal	Fazoviy diagonal	Пространственная диагональ
Perpendicular section	Perpendikulyar kesim	Перпендикулярное сечение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A right prism has lateral faces that are:

parallelograms

rectangles

triangles

trapezia

A section parallel to the base is:

smaller

congruent to the base

a triangle

a point

The space diagonal of an $a \times b \times c$ cuboid is:

$a + b + c$

$\sqrt{a^2 + b^2}$

$\sqrt{a^2 + b^2 + c^2}$

abc

The diagonals of a parallelepiped:

are equal

bisect each other

are perpendicular

do not meet

A cube of edge 5 has space diagonal:

$5\sqrt{2}$

$5\sqrt{3}$

15

25

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Space diagonal of a $3 \times 4 \times 12$ cuboid

ANSWER 13

2. Space diagonal of a cube of edge 5

ANSWER $5\sqrt{3}$

3. Face diagonal of a cube of edge 5

ANSWER $5\sqrt{2}$

4. Lateral faces of a right prism are

ANSWER rectangles

5. Section parallel to the base is

ANSWER congruent to the base

6. How many space diagonals has a cuboid?

ANSWER 4

7. Space diagonal of a $1 \times 2 \times 2$ cuboid

ANSWER 3

Medium · the standard question

7 PROBLEMS

1. Space diagonal of a $6 \times 8 \times 10$ cuboid

ANSWER $10\sqrt{2}$

2. Cube with space diagonal $9\sqrt{3}$; find the edge

ANSWER 9

3. Regular hexagonal prism, base edge 4, height 9: longest diagonal

ANSWER $\sqrt{145}$

4. Cuboid $2 \times 3 \times 6$; space diagonal

ANSWER 7

5. A cuboid has square base 5 and diagonal 13. Find the height

ANSWER $\sqrt{119} \approx 10.9$

6. Diagonal section of a cube of edge 6: its area

ANSWER $36\sqrt{2}$

7. Perpendicular section of a right prism is

ANSWER congruent to the base

Hard · stretch and reason

7 PROBLEMS

1. A cuboid has edges in ratio $1:2:3$ and diagonal $\sqrt{56}$. Find them

ANSWER 2, 4, 6

2. Prove the four space diagonals of a parallelepiped meet at one point

ANSWER Each diagonal section is a parallelogram

3. Oblique prism, lateral edge 10, perpendicular-section perimeter 18. Lateral area

ANSWER 180

4. A cube of edge a : angle between the space diagonal and a face diagonal

ANSWER $\approx 35.26^\circ$

5. A cuboid with all four space diagonals equal must be

ANSWER any cuboid — they always are

6. A parallelepiped whose diagonals are all equal is

ANSWER a cuboid

7. A regular triangular prism, base edge 6, height 8: longest diagonal

ANSWER 10

07 Homework

Homework — 5 tasks

Every diagonal answer must show both Pythagoras steps.

- Find the space diagonal of a $5 \times 12 \times 84$ cuboid.
- A cube has space diagonal 12. Find its edge and its face diagonal.
- A regular hexagonal prism has base edge 5 and height 12. Find its longest diagonal.
- Prove that the four space diagonals of a cuboid are equal.
- A cuboid has edges in the ratio $2 : 3 : 6$ and space diagonal 14. Find the edges.